EXPLORATORY DATA ANALYSIS ON USED CARS

Submitted by: Inchara Yadav
Department: Computer Science and Artificial Intelligence Engineering
Semester: Third Semester
Batch: ML-september
Date: 25-10-2026

Dataset Source: Kaggle

---

1. OBJECTIVE

The main objective of this project is to perform Exploratory Data Analysis (EDA) on a used cars dataset using Python.

The project aims to understand the structure of the data, identify missing values and duplicate records, clean the dataset, calculate descriptive statistics, and visualise patterns in car prices and other attributes.

The analysis focuses on the following questions:

1.Which fuel type is most common in the dataset?

Answer:-petrol

2.How are used car prices distributed?

Answer:- The selling prices range from ₹2,60,000 to ₹21,00,000. The average selling price is ₹7,59,600, while the median selling price is ₹6,60,000. The average is higher than the median because some cars have comparatively high selling prices.

3.How does manufacturing year relate to selling price?

Answer:-Newer Cars Generally Have Higher Prices.

4.How does distance travelled relate to selling price?

Answer:-Kilometres Driven Has a Negative Relationship with Price

5.Do prices differ between fuel types and transmission categories?

Answer:- Automatic Cars Have a Higher Average Selling Price.

2. DATASET OVERVIEW

The dataset contains information about used cars and their characteristics. The selected variables are used to understand differences in car prices and explore possible relationships between vehicle attributes.

The notebook previously reviewed used a 50-row illustrative dataset with seven columns. Confirm these details against the final dataset before submission.

Dataset Variables

Variable	Description
Car_Name	Name of the car
Year	Manufacturing year
Selling_Price_INR	Selling price in Indian rupees
Kms_Driven	Distance travelled by the car
Fuel_Type	Fuel category
Transmission	Transmission category
Owner	Ownership category

The dataset was loaded into a Pandas DataFrame for inspection and analysis. Functions such as head(), shape, info() and describe() were used to understand the records, column names, data types and statistical summaries.

Dataset size: Enter the confirmed number of rows and columns from your final notebook.

3. DATA QUALITY AND CLEANING

Data cleaning was performed to improve the consistency of the dataset and prepare it for analysis.

3.1 Missing Values

Missing values were checked using the isnull().sum() function. The results were reviewed to determine whether any values required treatment.

Observation: Record the actual missing-value counts from your notebook.

3.2 Duplicate Records

Duplicate rows were checked using the duplicated().sum() function. Duplicate records, if present, were reviewed and removed where appropriate.

Observation: Record the actual number of duplicate rows found before cleaning.

3.3 Data Type and Category Checking

Column data types and categorical values were inspected. Extra spaces and inconsistent text formatting were checked to maintain consistency.

Numerical columns were checked to ensure that they could be used for statistical analysis.

3.4 Outlier Detection

Box plots and the interquartile range (IQR) method were used to investigate potential outliers. Unusual values were checked before removal because they may represent genuine observations rather than errors.

3.5 Cleaning Summary

The original and cleaned dataset sizes should be compared in the final report.

- Original number of records: 50
- Duplicate records removed:0 duplicate values
- Missing values handled: 4
- Outlier records removed, if justified: 1

4. DESCRIPTIVE ANALYSIS

Descriptive analysis was performed to summarise the numerical and categorical variables.

The describe() function was used to calculate statistics such as count, mean, standard deviation, minimum, median and maximum.

For numerical variables, the analysis helps explain the range and variation in manufacturing year, selling price and kilometres driven.

For categorical variables, frequency counts help identify common fuel types, transmission categories and ownership categories.

Observation: Insert the actual descriptive-statistics table and category counts generated by your notebook. Explain the important values shown.

5. UNIVARIATE, BIVARIATE AND MULTIVARIATE ANALYSIS

5.1 Univariate Analysis

Univariate analysis examines one variable at a time. Histograms and count plots were used to understand numerical distributions and categorical frequencies.

5.2 Bivariate Analysis

Bivariate analysis examines relationships between two variables. Scatter plots and box plots were used to compare selling price with manufacturing year, kilometres driven, fuel type and transmission category.

5.3 Multivariate Analysis

Multivariate analysis examines three or more variables together. A correlation heatmap was used to examine numerical relationships. Multivariate scatter plots can also compare selling price and distance travelled while distinguishing categories such as fuel type or transmission.

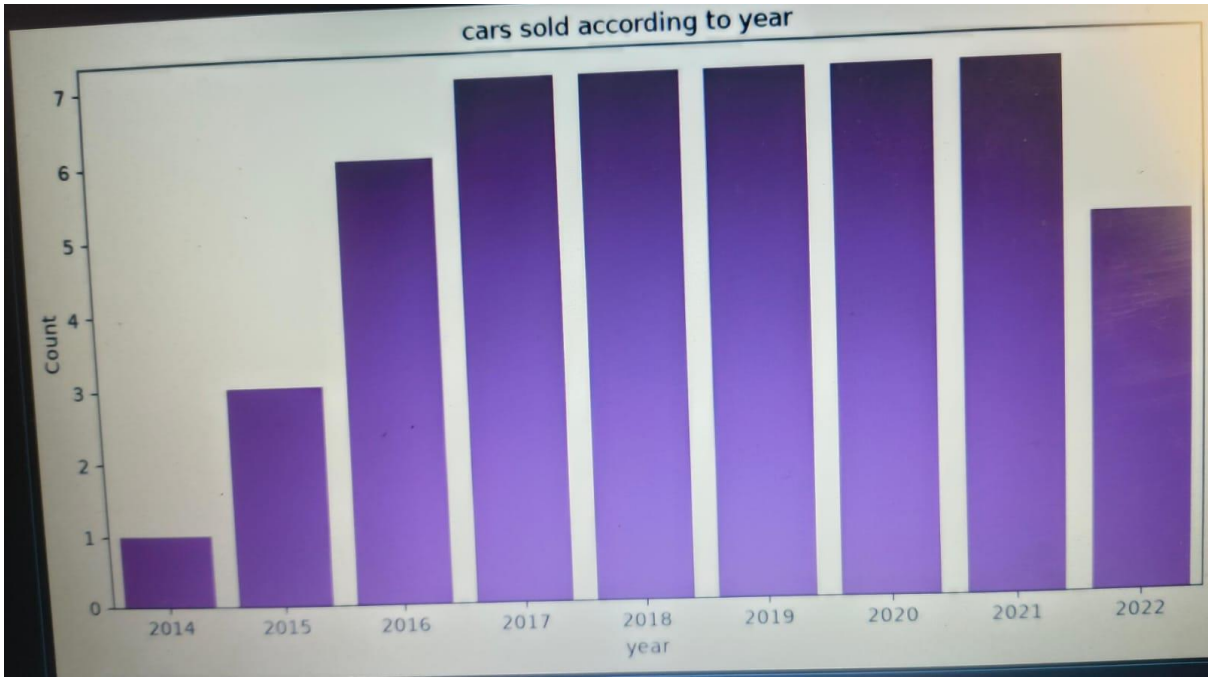
These techniques help identify patterns and associations within the dataset. They do not, by themselves, establish cause and effect.

6. VISUAL EDA AND OBSERVATIONS

Graph 1: Cars Sold According to Year

Insight 1: Distribution of Cars by Year

The dataset contains cars manufactured between 2014 and 2022. The count plot shows that 2021 has the highest number of cars, while 2014 has the lowest. The number of cars is relatively high from 2017 to 2021, indicating that these years are well represented in the dataset.



Graph 2: Year vs Selling Price

Insight 2: Manufacturing Year and Selling Price

The bar graph indicates a general increase in average selling price for newer cars. Cars manufactured in 2021 and 2022 have higher average selling prices than older cars. The correlation coefficient between Year and Selling_Price_INR is approximately +0.47, supporting a moderate

positive relationship.

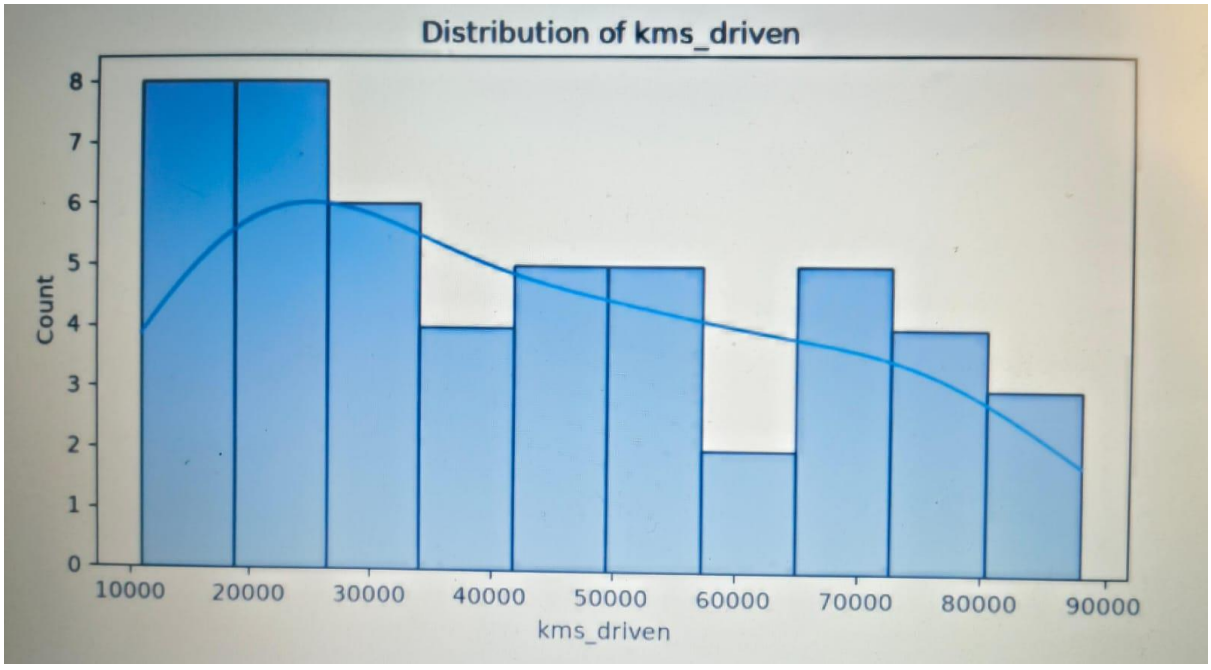


Graph 3: Distribution of Kilometres Driven

Insight 3: Distance Travelled and Selling Price

The histogram shows that many cars have travelled around 10,000 to 35,000 kilometres. The correlation between Kms_Driven and Selling_Price_INR is approximately -0.32.

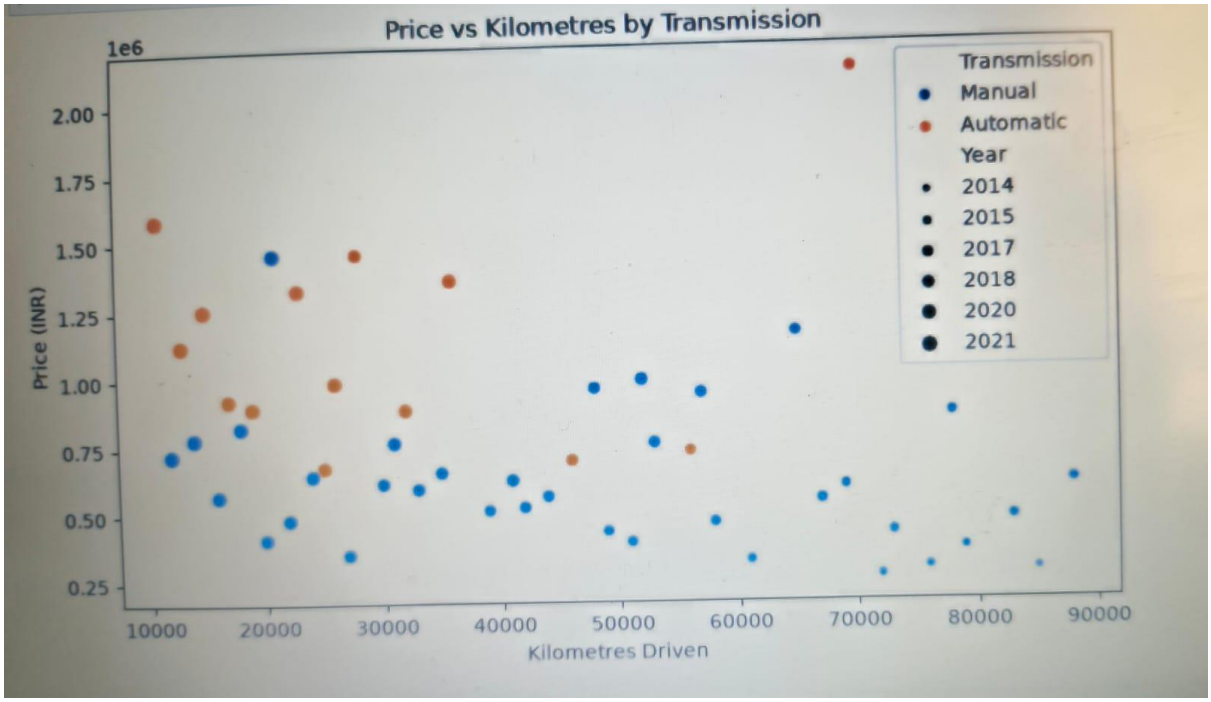
This indicates a weak-to-moderate negative relationship, suggesting that cars with higher mileage may tend to sell for lower prices.



Graph 4: Price vs Kilometres by Transmission

Insight 4: Relationship Between Year and Kilometres Driven

The heatmap shows a correlation of approximately -0.95 between Year and Kms_Driven. This is a very strong negative correlation, meaning newer cars in this dataset generally have lower recorded kilometres driven than older cars.

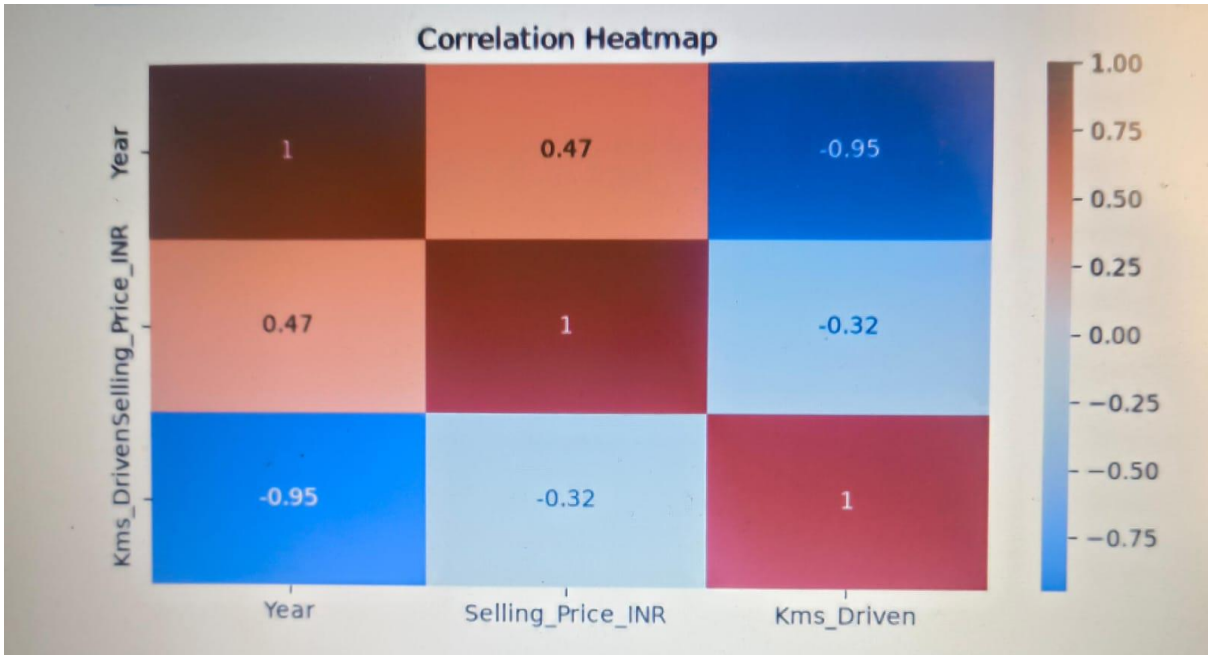


Graph 5: Correlation Heatmap

Insight 5: Selling-Price Variation and Potential Outliers

The box plot shows a wide range of selling prices and a possible high-price outlier in the initial graph. This indicates that some cars are priced considerably higher than most others. Such values should be investigated before removal because they may represent expensive vehicles rather than

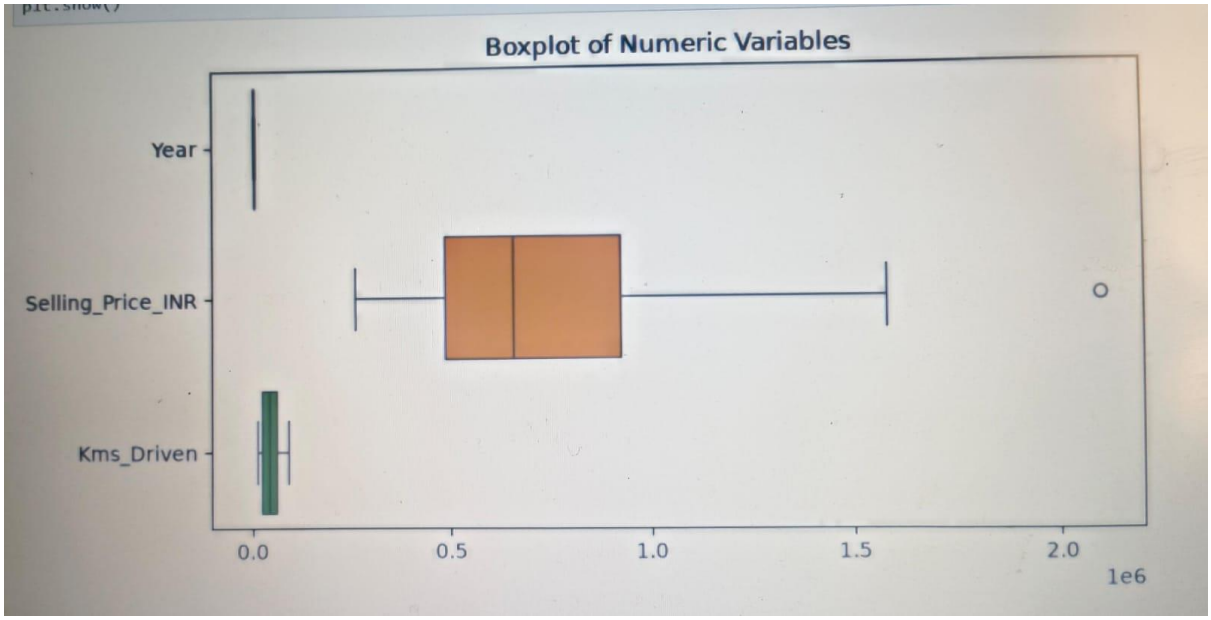
incorrect data.



Graph 6: Box Plot of Numeric Variables

Insight 6: Transmission and Selling Price

The scatter plot shows that both manual and automatic cars appear at different price levels. Some automatic cars have relatively high selling prices, but the graph alone does not prove that automatic transmission causes a higher price. Other factors, such as manufacturing year, car model and kilometres driven, may also affect selling price.



7. KEY FINDINGS AND INSIGHTS

The following are the five main questions investigated in the project. Final statements should be completed using the actual numerical results and graph patterns.

Insight 1: Fuel Type Distribution

The count plot was used to identify the most common fuel type. The category with the highest frequency in the actual graph should be reported here.

Finding: The most common fuel type is petrol, with 32 records.

Insight 2: Selling Price Distribution

The histogram and descriptive statistics were used to examine the distribution of selling prices.

Finding: The observed price range is ₹2,60,000 to ₹21,00,000.

Insight 3: Manufacturing Year and Price

The relationship between manufacturing year and selling price was examined using a graph or grouped average prices.

Finding: The data shows 0.47 between manufacturing year and selling price.

Insight 4: Distance Travelled and Price

A scatter plot was used to examine whether kilometres driven are associated with selling price.

Finding: The relationship appears -0.32(positive, negative or weak) based on the plotted data.

Insight 5: Correlation Between Numerical Variables

The correlation heatmap was used to identify numerical variables with stronger linear relationships.

Finding: The strongest observed correlation is -0.95.

8. LIMITATIONS

The findings are limited by the size, quality and characteristics of the selected dataset. If the illustrative 50-row dataset is used, it should not be treated as verified real-world used-car sales data.

The analysis may not account for important factors such as vehicle condition, service history, accident history, local demand and market conditions.

Correlation does not prove causation. Unusual values also require careful investigation before they are removed.

A larger, publicly sourced dataset with documented origins would provide a stronger basis for conclusions about the used-car market.

9. CONCLUSION

This project demonstrates the use of Python for Exploratory Data Analysis on used cars. The workflow includes dataset inspection, data cleaning, descriptive statistics, univariate analysis, bivariate analysis, multivariate analysis and visualisation.

Different charts were used to understand the distribution of selling prices, compare categories, examine relationships between variables and investigate potential outliers.

The project provides practical experience in handling data, creating visualisations and communicating findings. Future work could use a larger public dataset and examine additional vehicle characteristics to obtain more reliable and detailed insights.

10. FUTURE SCOPE

Future improvements may include:

- Using a larger and verifiable public used-car dataset.
- Including additional variables such as engine capacity, mileage and vehicle condition.
- Comparing prices across more brands and locations.
- Investigating unusual records in grea